

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

Bouke Klein Teeselink^{*1,2,3}

¹*Department of Political Economy, King's College London*

²*AI Objectives Institute*

³*King's Institute for Artificial Intelligence*

December 21, 2025

Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

^{*}Contact: bouke.klein.teeselink@kcl.ac.uk. I would like to thank Philip Tomei, Barbara Piotrowska, Bharat Chandar, and seminar participants of the Economics of TAI group for insightful comments. I particularly want to thank Jacob Schaal for exceptional research assistance.

ket characteristics. We examine heterogeneity along two dimensions: compensation levels (splitting firms and occupations by median pre-treatment salary) and geographic concentration (comparing London-based versus non-London entities). These dimensions are highly policy-relevant, as compensation-level heterogeneity provides a first indication as to whether LLMs may compress or widen wage inequality, while geographic variation tests whether disruption concentrates in London’s knowledge economy or extends throughout the UK labor market.

4.3.1 Heterogeneity by Compensation Level

Table A2 examines how firm-level employment effects vary by average compensation, with high-compensation jobs serving as a proxy for more highly educated white collar work, and low-compensation jobs for relatively low-educated blue collar work. We split firms at the median pre-treatment compensation level. Panel A shows that high-compensation firms experience substantially larger employment reductions than their lower-paying counterparts. A one standard deviation increase in LLM exposure reduces total employment at high-compensation firms by 0.7 percent ($p = 0.009$), driven primarily by a 0.8 percent reduction in low-seniority positions ($p = 0.005$). In contrast, Panel B indicates that low-compensation firms show no significant employment effects, with point estimates near zero for both total employment and low-seniority positions.

This pattern extends to hiring intentions. Table A4 shows that a one standard deviation increase in LLM exposure reduces the probability of posting vacancies by 1.3 percentage points at high-compensation firms ($p < 0.001$), compared to a 0.7 percentage point reduction at low-compensation firms ($p = 0.001$). While both groups curtail hiring, the effect is approximately twice as large at high-compensation firms.

The occupation-level analysis in Table A5 provides further evidence that the employment effects are concentrated among the highly paid. In high-salary occupations, a one standard deviation increase in LLM exposure is associated with a 5.7 percent reduction in job postings ($p = 0.009$) and £782 salary declines ($p < 0.001$), while low-salary occupations show no significant changes in posting volumes and minimal salary effects.

These results suggest that LLM disruption is almost entirely driven by changes

Table A6: Effect of LLM exposure on occupation-level job outcomes, London vs. non-London

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: Job in London				
LLM exposure x Post	-0.034** (0.017)	-562.818*** (139.355)	-0.010*** (0.002)	74.609 (132.525)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,734	16,734	16,734	16,625
Adjusted R ²	0.954	0.922	0.926	0.605
Panel B: Job outside London				
LLM exposure x Post	-0.040** (0.018)	-475.947*** (141.444)	-0.011*** (0.003)	-119.267 (111.131)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,812	16,812	16,812	16,799
Adjusted R ²	0.961	0.938	0.945	0.683

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for London-based jobs and jobs based outside London. Panel A presents results for job openings in London, Panel B for job openings outside London. All other definitions are as in Table 4.

A2.3 Occupation-Level Job Listings

Table A5: Effect of LLM exposure on occupation-level job outcomes, high vs. low salary occupations

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: High salary occupations				
LLM exposure x Post	-0.057*** (0.022)	-782.371*** (184.426)	-0.013*** (0.003)	-354.844*** (131.069)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,413	8,413	8,413	8,412
Adjusted R ²	0.964	0.903	0.901	0.726
Panel B: Low salary occupations				
LLM exposure x Post	-0.020 (0.031)	-112.614 (179.137)	-0.004 (0.004)	240.391 (170.804)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,402	8,402	8,402	8,392
Adjusted R ²	0.961	0.817	0.860	0.436

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for high- and low-salary occupations. Occupations are split at the median pre-treatment average posted salary (October 2022). Panel A presents results for occupations above the median salary, Panel B for occupations below the median. All other definitions are as in Table 4.